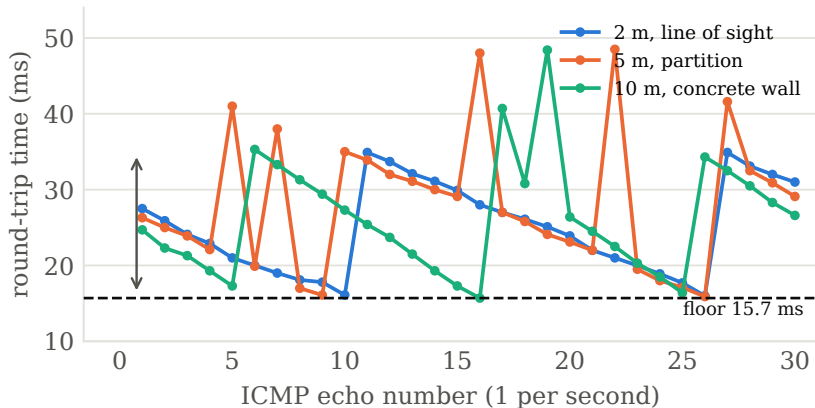
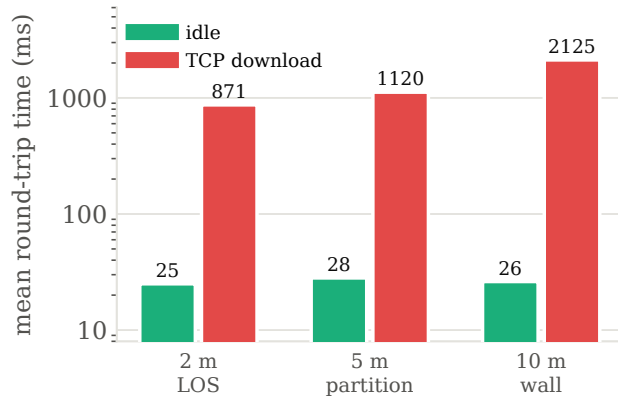


The testbed's latency is fine until the 6.17 MB downlink queue fills

Unloaded: a 16 ms floor plus a sawtooth



...and what a saturating download does to it



Left: the arrow marks the ≈ 19 ms window the round-trip time sweeps. The 1 s ping interval drifts 1.2 ms per packet against a periodic uplink scheduling opportunity, so each echo waits a different fraction of that cycle. Distance changes the floor by 0.3 ms; the queue changes the mean by 35–81 $\times$.